MARCO DE NEGRI

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RESEARCH

ETH AI Center (Apertus group) **Zürich, Switzerland**
Degeneration Detection Is Not Early Warning – accepted at NeurIPS 2026 Workshop Spring 2026

- Trained linear probes on the hidden states of Apertus-8B, Llama-3.1-8B and Mistral-7B to warn, token by token, that a model is about to fall into a repetitive loop, before it shows in the text.
- Proposed ONWARD, an evaluation that fixes the false-alarm rate on healthy responses and scores alarms against a human-verified failure onset; showed that AUROC saturates on this task and cannot tell an early warning from a late detection, and that LoRA adaptation below the probe more than doubles warning coverage on all three models. Run on the Alps supercomputer (CSCS).

Learning & Adaptive Systems Group (Prof. Andreas Krause, Scott Sussex) **Zürich, Switzerland**
Generative Antibody Design with Protein Language Models Spring 2026 – present

- Turning ESM2, a masked protein language model, into a generative model for antibody design, fine-tuning on yeast-display binding data via Direct Preference Optimization (DPO) to generate higher-affinity CDR-H3 variants of the C05 broadly neutralizing anti-flu antibody.
- Showed that evotuning (unsupervised fine-tuning on antibody sequence databases) gives a better starting point for DPO than the vanilla model, especially in the low-data regime, reaching a Spearman correlation of 0.65 between model scores and experimental binding on held-out test sets.
- Generated novel variants via Gibbs sampling and beam search on the ETH Euler HPC cluster.

Computer Vision and Geometry Lab, ETH Zürich (Daniel Barath) **Zürich, Switzerland**
Goal-driven 3D Map Generation with Reinforcement Learning | Project Website Spring 2026

- Trained an RL agent using PPO to dynamically allocate resolution in a 3D map, keeping high detail only where relevant to the task under a tight memory budget, and built a supporting Python library with online adaptive resolution, integrated with the PEANUT navigation agent in the AI Habitat simulator, cutting memory usage by 88.6% in test episodes while maintaining task success.

PUBLICATIONS

- **M. De Negri**, L. Sartori, L. Baggi, M. Reihs, E. F. Ďurech, M. El-Assady, V. Pyatkin, A. Hedström. *Degeneration Detection Is Not Early Warning.* NeurIPS 2026 Workshop: Interpretability as a Science.
- Project Apertus (**M. De Negri**, contributor, Interpretability team). *From Open Text Models to Open Multimodal Reasoners: Apertus v1.5 Technical Report.* arXiv:2610.15542, 2026.

EDUCATION

ETH Zürich **Zürich, Switzerland**
Master of Science in Computer Science September 2025 - present
Major in Machine Intelligence **GPA: 5.63/6.0**

Politecnico di Milano **Milan, Italy**
Bachelor's in Engineering of Computer Systems September 2022 - July 2025
Final Grade: 110/110 with honors

- **Honors:** "Best Freshmen Youth Fund" issued by Politecnico di Milano.
- Tuition tax exemption for 2 years (merit-based).

Universidad de Málaga **Malaga, Spain**
Erasmus+ September 2024 - February 2025

University of California, Santa Barbara (UCSB) **Santa Barbara, California**
Summer Session in Communication August 2023

SKILLS

- **Programming Languages:** Python, Java, C/C++, JavaScript/TypeScript, SQL, LaTeX
- **ML & Data:** PyTorch, HuggingFace Transformers, NumPy, pandas, scikit-learn, Matplotlib, W&B
- **Tools & Infrastructure:** Git, Linux, SLURM (Euler HPC), Docker, React, Next.js

LANGUAGES

- Italian - Native | English - C1 (IELTS, Band 7 - July 2024)